Introductory Mathematical Economics

Prelims - Function

Delhi School of Economics

Prelims - Function

Reading

Reading: Simon and Blume 13.5

Function

Take two sets A and B.

A **function** f is a rule that maps each element of A to exactly one element in B.

That is $f(a) \in B \, \forall a \in A$.

We denote this by $f : A \longrightarrow B$.

Relation

Let S and T be two non-empty sets. Intuitively a relation ρ between S and T is a *rule* that associates some or all elements of S with elements of T

Definition

Let S and T be two non-empty sets. A relation ρ between S and T is a subset of $S \times T$. If ordered pair $(s, t) \in \rho$, then the element s of the set S is said to be related to t in T by the relation ρ.

If $(s, t) \in (S \times T) - \rho$ then s is said to be not related to t by the relation ρ.

Example:

Let S=$\{2, 3, 4, 5\}$, T=$\{11, 12, 13, 14\}$. A relation ρ between S and T is defined by specifying that an element s in S is related to an element t in T if s is a divisor of t, Then,

$\rho = \{(2, 12), (2, 14), (3, 12), (4, 12)\}$.

$(2, 11) \in S \times T$ but $(2, 11) \notin \rho$ since 2 is not a divisor of 11.

Binary Relation

A relation ρ between S and S is said to be a *binary relation* on S

Definition

Let S be a non-empty set. A binary relation ρ on S is a subset of the cartesian product $S \times S$.

If (a,b) is an element of $S \times S$ and $(a, b) \in \rho$ then a is said to be related to b by the relation ρ. This is expressed as $a\rho b$.

Example:

Let ρ be the subset of $Z \times Z$ given by $\rho = \{(x, y) \in Z \times Z | \text{x+y is even}\}$. Then ρ is a relation on Z defined by "$x\rho y$ if and only if x+y is even" for $x, y \in Z$. Here, (1,3)$\in \rho$ and (2,3)$\notin \rho$.

Mapping

- Let A and B be two non-empty sets. If f is a relation between A and B then an element x of A may be related to one element or no element or many elements of B by the relation f.
- A relation f with the property that each element x of A is related to exactly one element y of B is said to be a mapping from A to B.

Definition

Let A and B be two non-empty sets. A mapping f from A to B is a rule that assigns to each element x of A definite elements y in B

Mapping

Let S=$\{1, 2, 3, 4\}$, T=$\{a, b, c, d\}$. Let us examine the relation f_1, f_2, f_3, f_4 between S and T.

1. $f_1 = \{(1, a), (1, b), (2, c), (3, c), (4, d)\}$
2. $f_2 = \{(1, a), (2, b), (3, c)\}$
3. $f_3 = \{(1, b), (2, c), (3, d), (4, a)\}$
4. $f_4 = \{(1, b), (2, c), (3, d), (4, a)\}$

$f_1 \& f_2$ are not mapping whereas, $f_3 \& f_4$ are mapping.

Examples

Example 1 $f : A \longrightarrow B$
$A = \{\text{Bread, Butter, Milk, Meat, Chapati}\}$, $B = \{\text{Lunch, Breakfast}\}$

$$f(x) = \begin{cases} \text{Breakfast} & \text{if } x = \text{Bread, Butter, Milk} \\ \text{Lunch} & \text{if } x = \text{Meat, Chapati} \end{cases}$$

f is a function because each input in A has a single output in B. Note that many inputs have the same output, which is not a violation.

Example 2 $f : \mathcal{R} \longrightarrow \mathcal{R}^2$
$\mathcal{R}$ is the set of all real numbers. $\mathcal{R}^2$ is the set of all real vectors in 2-dimension

$$f(z) = \begin{pmatrix} -z \\ z \end{pmatrix}$$

This is also a function. The output set is the set of vectors and for each real number (input), we get one vector as output.

Example 3 $f : \mathcal{R} \longrightarrow \mathcal{R}$

$$f(x) = (-x, x), \text{ where } (-x, x) \text{ is an open interval in } \mathcal{R}$$

This is not a function because the output set is the set of real numbers, but we obtain multiple real numbers as output.

Domain and Range

$f : A \longrightarrow B$.

A is called the **domain** of f

$f(A) = \{b \in B \mid f(a) = b, \quad \text{for some } a \in A\}$ is the **range** of f

$a \in A$ is called a pre-image and $f(a)$ is called an image.

Inverse Map

Inverse of f, denoted by f^{-1}, is the route back from B to A.

For each $b \in f(A)$, $f^{-1}(b) = \{a \in A \mid f(a) = b\}$

f^{-1} is not necessarily a function

Inverse Map: Example

Let us identify the inverse of Example 1 and Example 2

Example 1 $f^{-1} : B \longrightarrow A$
$B = \{\text{Lunch, Breakfast}\}, A = \{\text{Bread, Butter, Milk, Meat, Chapati}\}$

$$f^{-1}(x) = \begin{cases} \{Bread, Butter, Milk\} & \text{if } x = \text{Breakfast} \\ \{\text{Meat, Chapati}\} & \text{if } x = \text{Lunch} \end{cases}$$

This is not a function as one input is mapped to many outputs.

Example 2 $f^{-1} : \mathcal{R}^2 \longrightarrow \mathcal{R}$

$$f^{-1}\begin{pmatrix} -z \\ z \end{pmatrix} = z$$

This is not a function because there are inputs that have no output. For example, $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$ has no output.

Composition of Functions

$f : A \longrightarrow B$ and $g : B \longrightarrow C$

$g \circ f$ is the composite function which maps each element of A to exactly one element in C, through a temporary stop at B.

$g \circ f(a) = g(f(a)) \ \forall a \in A$.

Results

$f : A \longrightarrow B$. $B_k \subseteq B$ for all $k \in \Lambda$
$f^{-1}(B_k) = \{z \in A | f(z) \in B_k\}$

(i) $\bigcup_{k\in\Lambda} f^{-1}(B_k) = f^{-1}(\bigcup_{k\in\Lambda} B_k)$

(ii) $\bigcap_{k\in\Lambda} f^{-1}(B_k) = f^{-1}(\bigcap_{k\in\Lambda} B_k)$

Results

$f : A \longrightarrow B$. $A_k \subseteq A$ for all $k \in \Lambda$
$f(A_k) = \{f(z) | z \in A_k\}$

(i) $\bigcup_{k\in\Lambda} f(A_k) = f(\bigcup_{k\in\Lambda} A_k)$

(ii) $\bigcap_{k\in\Lambda} f(A_k) \supseteq f(\bigcap_{k\in\Lambda} A_k)$

A Proof

We want to prove: $\bigcap_{k\in\Lambda} f(A_k) \supseteq f(\bigcap_{k\in\Lambda} A_k)$

Take $y \in f(\bigcap_{k\in\Lambda} A_k)$

$\Rightarrow y = f(z)$ where $z \in \bigcap_{k\in\Lambda} A_k$ (by definition of $f(\bigcap_{k\in\Lambda} A_k)$)

$\Rightarrow y = f(z)$ where $z \in A_k$ for all $k \in \Lambda$ (by definition of intersection)

$\Rightarrow y \in f(A_k)$ for all $k \in \Lambda$ (by definition of $f(A_k)$

$\Rightarrow y \in \bigcap_{k\in\Lambda} f(A_k)$ (Proved)

However $\bigcap_{k\in\Lambda} f(A_k) \subseteq f(\bigcap_{k\in\Lambda} A_k)$ is not true. Let us construct a counterexample.

$f : \mathcal{R} \longrightarrow \mathcal{R}$, $f(x) = |x|$.

$\Lambda = \{1, 2\}$. $A_1 = \{x | x \geq 0\}$ and $A_2 = \{x | x < 0\}$.

$f(A_1 \cap A_2) = \emptyset$ but $f(A_1) \cap f(A_2) = \{x | x > 0\}$

Types of Function

$f : A \longrightarrow B$ is **injective (one-to-one)** if $a \neq a'$ implies $f(a) \neq f(a')$

Example 1 is NOT one-to-one but *Example* 2 is one-to-one.

$f : A \longrightarrow B$ is **surjective (onto)** if for every $b \in B$ there exists $a \in A$ s.t. $b = f(a)$

Example 1 is onto but *Example* 2 is NOT onto.

A function that is both one-to-one and onto is called **bijective** Example: $f : \mathcal{R} \longrightarrow \mathcal{R}$; $f(x) = 2x$

Bijective Function

$f : A \longrightarrow B$ is bijective

- f^{-1} is a function from B to A
- f^{-1} is a bijective function
- $f^{-1}(f(a)) = a \, \forall a \in A$ and $f(f^{-1}(b)) = b \; \forall b \in B$.

Theorems

Theorem

If $f : A \longrightarrow B$ and $g : B \longrightarrow C$ are both surjective, then the composite function $gof : A \longrightarrow C$ is surjective.

Proof.

Let z be an element of C
Since g is surjective, there is at least one pre-image of z in B. Let one such be y. Then $y \in B$ and g(y)=z.
Since f is surjective and $y \in B$, there is at least one pre-image of y in A. Let one such be x. Then $x \in A$ and f(x)=y.
gof(x)=g(y)=z.
This implies that z has a pre-image in A under the mapping *gof*. Since z is arbitrary, *gof* is surjective. □

Theorems

Theorem

If $f : A \longrightarrow B$ and $g : B \longrightarrow C$ be two functions such that $gof : A \longrightarrow C$ is injective then f is injective.

Proof.

If possible, let f be not injective. Then there exists two distinct elements x_1, x_2 in A such that $f(x_1) = f(x_2)$.
So $gof(x_1) = gof(x_2)$ and this contradicts that gof is injective.
Therefore, f is injective. □

Theorems

Question: If $f : A \longrightarrow B$ and $g : B \longrightarrow C$ be two functions such that $gof : A \longrightarrow C$ is injective. Is g injective?

Let's take a counter-example,

Let $f : R \longrightarrow R$ be defined by $f(x) = e^x, x \in R$ and $g : R \longrightarrow R$ be defined by $g(x) = x^2, x \in R$.

Here, $gof : R \longrightarrow R$ is defined by $gof(x) = e^{2x}, x \in R$

gof is injective, but g is not injective.